



College of Computing and Software Engineering

CS 3305: Data Structures

Course Section: W01 (83201)

Fall Semester 2026 Syllabus

Instructor Information

Maxwell Bradley

Email: mbradl59@kennesaw.edu

Office Location: Norton Hall (R2) 329

Office Phone: mbradl59@kennesaw.edu

Office Hours: TBD (Office, drop-in) Virtual, by appointment (flexible)

Preferred Method of Communication: Email

Course Information

Total Credit Hours: 3

Prerequisite: (Math 2345 or CSE 2300) and [(CSE 1322 and CSE 1322L each with a "B" or better), or MTRE 2610 with a "B" or better, or CPE 3000 with a "B" or better]

Course Meeting Information

Course Meeting Time: Asynchronous

Course Meeting Location: D2L/MS Teams

Course Meeting Modality: Online

Additional Course Information: Recorded lectures from past course offerings are available on D2L. Lectures from current course offerings (if any) will be streamed and recorded on MS Teams.

Course Materials

Introduction to Java Programming and Data Structures

ISBN: 9781292420059

Authors: Y. Daniel Liang

Publisher: Pearson

Publication Date: 2021-01-01

Edition: 10th or later

Recommended

Course Description

This course introduces data structures, specification, application, and implementation. The case studies will illustrate how data structures are used in computing applications. The emphasis of the course is on linear and some nonlinear data structures and object oriented principles. Topics include: abstract data types, stacks, queues, lists, binary search trees, priority queues, recursion, algorithm efficiency, trees, heaps, hash tables, and analysis of search and sort algorithms and their performance for implementation and manipulation. The programming language to be used in this course is any standard high-level object-oriented programming language such as C++, Java, and Ada.

Course Learning Outcomes

- Demonstrate understanding of single dimensional, multidimensional arrays, and dynamic arrays to store and access data, by applying these to programs
- Demonstrate learning of specifications and presentation of commonly used data structures by completely developing software. This must include appropriate documentation
- Demonstrate understanding and learn of advanced search and sort algorithms and their performance issues, by applying these to programs

- Demonstrate the analysis of the time complexity and space complexity of algorithms by explaining these concepts and applying these in programming
- Demonstrate the data structure covered in the courses by performing problem solving and application development
- There are no course materials required for this course.

Course Requirements and Assignments

This course consists of 9 programming assignments, 15 weekly coding challenges, and 2 exams. Information on assignments and exams can be found in D2L and in the course schedule.

Programming assignments are typically due by 11:59PM on Wednesdays. Grading policy for the programming assignments is strictly by the rubric. You should consult the Assignment Grading Guidelines as well as the rubric for each assignment carefully prior to submitting each assignment. **It is very easy to lose 50 or 100 points for small, easily avoidable errors if you are not careful.**

Coding challenges are typically due by 11:59PM on Fridays. Grading policy for the coding challenges is forgiving and based on honest effort. When completing coding challenges, you should prioritize your own learning, whatever that looks like for you.

Exams are administered asynchronously using LockDown Browser with Webcam. You can take them whenever you want within the allotted time period (varies from 24 hours to about a week).

Evaluation and Grading Information

Grading Scale:

A 89.5% - 100%

B 79.5% - 89.49%

C 69.5% - 79.49%

D 59.5% - 69.49%

F 0% - 59.49%

Assignment Weights:

40% Programming Assignments (x9)

20% Weekly Coding Challenges (x15)

20% Midterm Exam

20% Final Exam

Grade Disputes:

There are no makeup exams, alternative assignments, or extra credit opportunities.

Assignment grading issues/disputes must be resolved with the instructor (not the GTA) within 10 days from the date the assignment grade is published in D2L.

There will be no change to the grade after 10 days.

Late Submissions:

No late submissions will be accepted. Assignment due dates can be found in D2L. Submissions flagged as late receive an automatic zero. Note that D2L will flag your submission late if you submit it at 11:59:01pm.

Early Alert / Progress Reports

This class is participating in Early Alerts, which faculty will submit during the first few weeks of class. These reports notify advisors and student success coaches to a range of things like missing class, missing assignments, if you might need to visit tutoring, or could benefit from help with time management or study skills. These are not recorded as grades and are meant to provide you with some additional resources to ensure you can be successful in your class. If you receive notification of an early alert, please take advantage of these resources.

Midterm Grades

A midterm grade may be assigned by the midterm grade due date identified on the academic calendar. This midterm grade is for assessing mid-semester performance at least one week prior to the last day to withdraw without academic penalty. You may view your midterm grade in Owl Express. Note that only your final grade will be officially recorded on your academic transcript.

Course Calendar

This is a rough timeline for the content which will be covered in this course. Some topics may take more or less than 1 week to cover, which may cause the schedule to become misaligned. I reserve the right to adjust or modify the course and its assignments or their deadlines as necessary based on the particular demands of the semester. When changes are necessary, I will communicate the pertinent information via D2L announcements.

As a reminder, Programming Assignments are due on Wednesdays, and Coding Challenges are due on Fridays.

C1, the first coding challenge, is ungraded (free points) since it is due during the add/drop period. A submission is not necessary.

C15, the last coding challenge, is optional (replaces your lowest coding challenge) since it is due during finals week.

Course Calendar.

Week	Date (Mon)	Topic Focus	Assessments / Deliverables
1	Aug 24	Syllabus & Policies / OOP Review / Recursion (Ch-18)	Challenge #1
2	Aug 31	Generics (Ch-19)	Challenge #2, Assignment #1
3	Sep 07	NO MONDAY CLASSES Lists & Linked-Lists (Ch-20)	Challenge #3, Assignment #2
4	Sep 14	Stacks (Ch-20)	Challenge #4, Assignment #3
5	Sep 21	Queues (Ch-20)	Challenge #5, Assignment #4
6	Sep 28	Big O Notation (Ch-22)	Challenge #6, Assignment #5
7	Oct 05	Sorting Algorithms (Ch-23)	Challenge #7
8	Oct 12	Sorting Algorithms (Ch-23)	Challenge #8, EXAM 1 (Midterm)

Week	Date (Mon)	Topic Focus	Assessments / Deliverables
9	Oct 19	Trees (Ch-25)	Challenge #9
10	Oct 26	Binary Search Trees (Ch-25)	Challenge #10
11	Nov 02	Heaps & Heap Sort (Ch-23)	Challenge #11, Assignment #6
12	Nov 09	AVL Trees (Ch-26)	Challenge #12, Assignment #7
13	Nov 16	Hash Tables (Ch-27)	Challenge #13
14	Nov 23	FALL BREAK – No Classes	
15	Nov 30	Graphs (Ch-28)	Challenge #14, Assignment #8
16	Dec 07	Wrap-Up / Exam Review	Challenge #15, Assignment #9
17	Dec 08+	FINALS WEEK	EXAM 2 (Final)

Course Policies

Attendance Policy:

This is a fully asynchronous class. Students will need to exercise a certain level of discipline and remain diligent in keeping on track with the course timeline. The content is cumulative, meaning each lesson builds on the next, so falling behind will make it very difficult to get back on your feet. It is the student's responsibility to attempt each course activity at a regular pace and reach out for help when they need it.

Modified Operations Policy:

Kennesaw State University is committed to remaining open and continuing full operations, including class schedules, whenever possible. The Office of the President, in consultation with the KSU Office of Emergency Management and other officials, will determine when an emergency impacts KSU's normal operations. When the University is closed due to an emergency, all classes will be canceled, and non-essential personnel will not be expected to work.

When the University shifts to modified operations, Cabinet members, in

consultation with the Office of the President or designee and the Office of Emergency Management, will be responsible for determining appropriate operational continuity for their divisions. This may include remote work. During modified operations, each faculty member will be responsible for instructional continuity for their classes. To ensure instructional continuity, all Academic Colleges and Departments are expected to maintain and regularly update instructional continuity plans. All syllabi should also include an instructional continuity plan statement specific to that course.

During a weather event or other emergency, those commuting to campus should exercise their best judgment when determining travel options. If the University remains open and an employee elects not to report to work, the employee will be required to use their accrued vacation time.

The University will announce campus closures and delayed schedules in several ways. The cell phone number on file with the university will automatically receive KSU Alerts, so make sure your information in OwlExpress is accurate at all times. An email will also be sent to your university account.

In addition, announcements will be made by a notice on the Kennesaw State University home page.

Our class continuity plan includes:

Communication: Please check D2I Brightspace or e-mail for necessary instructions.

Virtual Classes: If in-person classes are not possible, we may transition to virtual classes using MS Teams.

Assignments and Assessments: Deadlines for assignments and assessments may be adjusted to accommodate the emergency situation.

We understand that emergencies create unique challenges. If you need additional support during an emergency, reach out via Brightspace or e-mail. The university also offers resources such as counseling and academic support, which can be accessed remotely.

AI Use Prohibited:

You are expected to generate your own work in this class. When you submit any kind of work, you are asserting that you have created it completely on your own unless you indicate otherwise using quotation marks and proper citation for the source(s) you used to help you. Submitting content that has been generated by someone other than you, or that was created or assisted by an AI generative tool is cheating and constitutes a violation of the KSU Code of Academic Integrity.

University Sponsored Academic Resources

University Academic Resources:

- [Academic Tutoring Services](#)
- [Disability Resources](#)
- [Library](#)
- [Supplemental Instruction](#)
- [The Writing Center](#)
- [Virtual Smart Center](#)
- [Academic Advising](#)

Student Resources:

- [Student Support Links for Online Learning](#)
- [Student Success Resources](#)
- [Scholarships and Financial Aid](#)
- [Office of the Registrar](#)
- [Office of the Bursar](#)

Grade Appeals and Student Complaints:

KSU desires to resolve student grievances, complaints and concerns in an expeditious, fair and amicable manner. The [complaints and concerns webpage](#) was developed to assist current and prospective students in submitting complaints and appeals, and to direct them to the most effective venue for accurate information and resolution. The resources on the page will direct students to the specific venue to appropriately address the related student complaint.

Complaints for online students are resolved following the same general procedures for students who attend classes on campus. However, for any

process that requires a student appear in person, the university may make other arrangements. For processes that cannot be completed via telephone, email, or written correspondence, the university may set up a two-way video-conference site in place of a meeting on the KSU campus.

Technology Resources

For issues with technical difficulties, please contact the Student Help Desk:

[Submit a Request for Help](#)

Email: service@kennesaw.edu

Call: 470-578-6999

[New Students Technology Services](#)- Steps that will help you meet the technological requirements of the University.

Any request for make-ups due to technical issues **MUST** be accompanied by the ticket number received from UITs when the problem was reported to them. The ticket number will document the time and date of the problem. You **MUST** e-mail your instructor within 24 hours of the technical difficulty if you wish to request a make-up.

Student Support and Wellness Resources

Student Support Resources

- [Department of Career Planning & Development](#)
- [Student Health Clinic](#)
- [Military and Veteran Services](#)

Student Wellness Resources

YOU MATTER at KSU, and your wellbeing is a priority to us. College is exciting, but it can also bring challenges that affect your health, wellbeing, and academic success. Whether you need academic accommodations through Student Disability Services, counseling,

health education, addiction and recovery support, fitness and recreation opportunities, or access to basic needs resources, [Wellbeing@KSU](#) is here to help you do well and be well.

Faculty Profile (USG CV Requirement):

Directory Information

Name: Maxwell Bradley

Title: Lecturer of Computer Science

Office Location: Norton Hall (R2), Room 329

Email: mbradl59@kennesaw.edu

Office Hours: MWF 11:00am - 12:00pm (Drop-in); Virtual by appointment.

Degrees Earned

M.S., Computer Science (Concentration: Machine Learning)

- *Georgia Institute of Technology*, December 2023. GPA: 4.00/4.00

B.S., Computer Science (Concentration: Media & Intelligence)

- *Georgia Institute of Technology*, December 2022. GPA: 3.95/4.00

Teaching Interests

Past Courses Taught:

- CS 3305 Data Structures
- CS 3410 Introduction to Database Systems
- CS 3503 Computer Organization and Architecture
- CS 4306 Algorithm Analysis
- CS 4308 Concepts of Programming Languages
- CS 4732 Machine Vision

Instructional Philosophy:

- My teaching often focuses on the fundamentals of computer science theory, which I frequently interrupt with comments regarding realistic modern industry standards.
- I value curricular agility. I view the classroom as a dynamic environment which pushes both students and teachers to learn relentlessly. I am committed to pedagogical experimentation, therefore I frequently adapt and modify my teaching methods to find the most effective strategy.

Institutional Policies

Academic Integrity Statement:

Every KSU student is responsible for upholding the provisions of the [Student Code of Conduct](#), as published in the Undergraduate and Graduate Catalogs. Section 5c of the Student Code of Conduct addresses the university's policy on academic honesty, including provisions regarding plagiarism and cheating, unauthorized access to university materials, misrepresentation/falsification of university records or academic work, malicious removal, retention, or destruction of library materials, malicious/intentional misuse of computer facilities and/or services, and misuse of student identification cards. Incidents of alleged academic misconduct will be handled through the established procedures of the Department of Student Conduct and Academic Integrity (SCAI), which includes either an "informal" resolution by a faculty member, resulting in a grade adjustment, or a formal hearing procedure, which may subject a student to the Code of Conduct's minimum one semester suspension requirement.

KSU Disruption of Campus Life policy:

All students are responsible for knowing the information, policies and procedures outlined in the Kennesaw State University Codes of Conduct. [Student Conduct and Academic Integrity \(SCAI\)](#) includes: the general Student Code of Conduct, the Residential Code of Conduct, and the Code of Academic Integrity.

KSU Web Accessibility:

Federal law Section 508 Subsection 1194.22 of the Rehabilitation Act and the Board of Regents (BOR) of the University System of Georgia (USG) Web Accessibility Guidelines require that all web content meet the federal government's accessibility guidelines. As such, KSU complies with USG guidelines.

University accessibility assistance is provided by several offices as noted below. Staff in these offices work to accommodate requests for access or assistance with access as soon as possible in order to either accommodate the request or identify an effective alternative for the requester.

Request for Office Contact Number Contact Email

- Student Support Services Student Disability Services 470-578-2666 sds@kennesaw.edu
- Student Technology Assistance [Student Technology Assistance](#)
- University Information Technology Services 470-578-6999 service@kennesaw.edu
- Third Party Technology Assistance University Information Technology Services 470-578-6999 service@kennesaw.edu

KSU Reasonable Accommodations:

Students with qualifying disabilities under the Americans with Disabilities Act (ADA) and/or Section 504 of the Rehabilitation Act who require "reasonable accommodation(s)" to complete the course may request those from Office of Student Disability Services. Students requiring such accommodations are required to work with the University's Office of Student Disability Services rather than engaging in this discussion with individual faculty members or academic departments. If, after reviewing the course syllabus, a student anticipates or should have anticipated a need for accommodation, he or she must submit documentation requesting an accommodation and permitting time for a determination prior to submitting assignments or taking course quizzes or exams. Students may not request retroactive accommodation for needs that were or should have been foreseeable. Students should contact the office as soon as

possible in the term for which they are seeking accommodations. Student Disability Services is located in Kennesaw Hall, Room 1205 on the Kennesaw campus and Building A in Suite 160G on the Marietta campus. Please visit the [Student Disability Services \(SDS\) website](#) for more information, or call the office at 470-578-2666 (Kennesaw campus) or 470-578-7361 (Marietta campus).

KSU Enrollment Management/Course Attendance:

Students are solely responsible for managing their enrollment status in a class; nonattendance does not constitute a withdrawal.

Copyright Law:

It is the responsibility of KSU faculty and students to respect the rights of copyright holders and complying with copyright law. The University System of Georgia recognizes that the exclusive rights of copyright holders are balanced by limitations on those rights under federal copyright law, including the right to make a fair use of copyrighted materials and the right to perform or display works in the course of face-to-face teaching activities.

The University System of Georgia facilitates compliance with copyright law and, where appropriate, the exercise in good faith of full fair use rights by faculty and staff in teaching, research, and service activities. The University System of Georgia ensure compliance with copyright law in the following ways.

1. The USG informs and educates students, faculty, and staff about copyright law, including the limited exclusive rights of copyright holders as set forth in 17 U.S.C. § 106, the application of the four fair use factors in 17 U.S.C. § 107, and other copyright exceptions.
2. The USG develops and makes available tools and resources for faculty and staff to assist in determining copyright status and ownership and determining whether use of a work in a specific situation would be a fair use and, therefore, not an infringement under copyright law;
3. The USG facilitates use of materials currently licensed by the University System of Georgia and provides information on licensing of third-party materials by the University System; and
4. The USG identifies individuals at the University System and member institutions who can counsel faculty and staff regarding application of copyright law.

Protecting Students' Privacy (FERPA):

Students have certain rights to privacy. These rights are mandated by federal policy. [The Student Handbook](#) contains information regarding Rights Pertaining to Student Records, and FERPA specific details are available on the [KSU FERPA website](#).

A key requirement of the formal evaluation process is the protection of individual privacy rights concerning educational grading. The University's online learning system and email system is designed to prevent unauthorized individuals from gaining access to sensitive information or information protected by federal or state law. Consequently, faculty and students are strongly encouraged to only communicate regarding course matters through the University's designated technology learning system.

Information should not be made public in any way in which a student's grades, social security number, or other personal information may be identified. Grade information may be shared with members of the KSU community who also have a legitimate educational interest in student success (e.g. academic advisors or members of the Behavioral Response Team). Faculty may be asked to provide early alert information if there is a concern that a student is at risk, academically or otherwise.

KSU Sexual Misconduct:

In accordance with federal and state law including, Title IX of the Education Amendments of 1972 ("Title IX") and Title VII of the Civil Rights Act of 1964 (Title VII), the University System of Georgia (USG), including Kennesaw State University, prohibits discrimination on the basis of sex in any of its education programs or activities or in employment. The USG is committed to ensuring the highest ethical conduct of the members of its community by promoting a safe learning and working environment. To that end, Kennesaw State University follows USG Board of Regents Policy Manual, Section 6.7. [Visit Title IX](#).

Withdrawal from Classes:

A student who officially withdraws from a course by the end of the last day to withdraw without academic penalty will receive a grade of “W” and receive no credit.

A student will receive a refund only when the student withdraws from ALL courses for the applicable semester and only by the schedule outlined in the University refund policy found in the Tuition, Expenses, & Financial Aid section of the undergraduate and graduate university catalogs.

Students should be aware that a reduction in their hours might result in the loss of full-time student status and thus affect their financial aid, scholarships, athletic and ticket eligibility, University housing accommodations, use of University resources and access to University facilities, visa for international students, and Veterans Educational Benefits. Students should contact the appropriate office and their academic advisor with questions about the impact of their withdrawal from a course before initiating a withdrawal. Veterans and dependents of veterans who receive educational benefits must notify the Veterans Education Benefits Area in the Office of the Registrar of any course load reductions.

Hardship Withdrawals

If a student experiences significant personal hardship (e.g., medical or family emergency, prolonged illness), the Dean of Students can approve a hardship withdrawal from all courses in the term for which the student is currently registered. In the case of an approved hardship withdrawal from all courses, the Registrar will assign grades of “W” for those classes. The deadline for final approval of a hardship withdrawal by the Dean of Students is the last day of class for which the hardship withdrawal is sought. If the hardship withdrawal process is not complete by the last day of class for which the hardship withdrawal is sought, a student must appeal for a retroactive hardship withdrawal from the Academic Standing Committee.

Appeals for retroactive hardship withdrawals must be directed to the Academic Standing Committee. Retroactive hardship withdrawals are rarely granted if it

has been more than one year since the last day of class for which the withdrawal is sought. Extraordinary justification must be shown. In the case of approved retroactive hardship withdrawals, the Registrar will assign a grade of “W.”

University Initiated Withdrawals

If a student is suspended by the Office of Student Conduct following a violation of the University’s Code of Conduct not related to academic dishonesty, the Office of Student Conduct may facilitate a University-initiated withdrawal from courses for which a student is registered for the term. The Registrar will assign grades of “W” for those classes.

Military Withdrawals

A student will receive a “WM” symbol for all courses and a full refund of tuition and mandatory fees and a pro rata refund of other fees for military and other service, as defined by [BOR Policy Manual, Section 7.3.5.3](#). To request a military withdrawal, the student must submit a copy of official orders to the Office of the Registrar.

Academic Feedback:

Institutional Chief Academic Officers will encourage faculty to clarify for students, at the beginning of each course, the basis on which grades will be determined and to provide timely academic feedback as the course progresses ([BOR Academic and Student Handbook Policy 2.18](#))

Netiquette: Communication Courtesy:

All members of the class are expected to follow rules of common courtesy in all email messages, threaded discussions and chats. See [Email Best Practices](#) for additional guidance.

Course Delivery:

KSU may shift the method of course delivery at any time during the semester in

compliance with University System of Georgia health and safety guidelines. In this case, alternate teaching modalities that may be adopted include hyflex, hybrid, synchronous online, or asynchronous online instruction. Please refer to the Instructional Continuity statement included in each course syllabus for specific plans.

Research Interests

Note: As a Lecturer, my primary focus is on instruction and curriculum development. However, I maintain academic and technical interests in the following areas:

As an occasional contributor to SLIAI under Dr. Simpkins:

- Machine Theory of Mind
- Interactive AI
- Offline Reinforcement Learning
- Value Alignment

Personal research:

- CNNs & ViTs
- Real-time video feed processing (classification/box regression)
- Infrastructure & crop damage detection; pest detection

Recent Scholarship

Co-advisor for the OWL Cyber Club; mentor to club's participation in the SVCC Cyberwarrior LLM Challenge (2025).

Advisor for the KSU Career & Technical Development Club.

Faculty mentor for the KSU Hackathon.